2017
COMPUTER SCIENCE — GENERAL
First Paper

(b) Find the value of $25_{10} - 35_{10}$ using 1's complement and 2's complement method.

(c) Find the dual of $F = X + YZ'$.

(d) Express $F = XY + X'Z$ in its canonical form.

(Digital Logic Design)

4. (a) Convert $(FACE)_{16}$ to its corresponding binary code.

(b) Explain how decoders can be used as demultiplexers.

(c) Show that multiplexers are functionally complete.

(d) Design a 3-input adder using two 2 input adders and logic gates of your choice. 3+4+5+4

5. (a) Design a 3-bit full sequence counter.

(b) Construct a 4-bit binary shift (bi-directional) register.

(c) Write down the characteristic expression of J-K flipflop. Show how a J/K Flipflop can be converted into a D flipflop. 5+5+(3+3)